BJT small-signal II

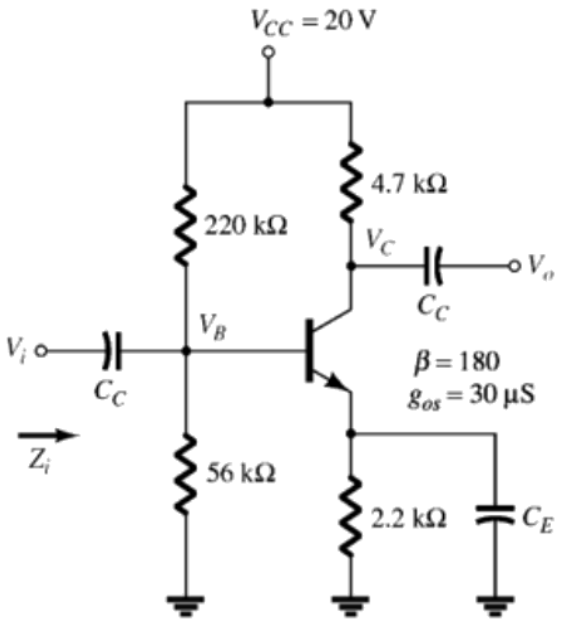
Practice questions · Day 12 · Wed 4 Nov · pack E7

5. Practice

From your course files

WS6 Q6 (course question). For the network of figure 6:

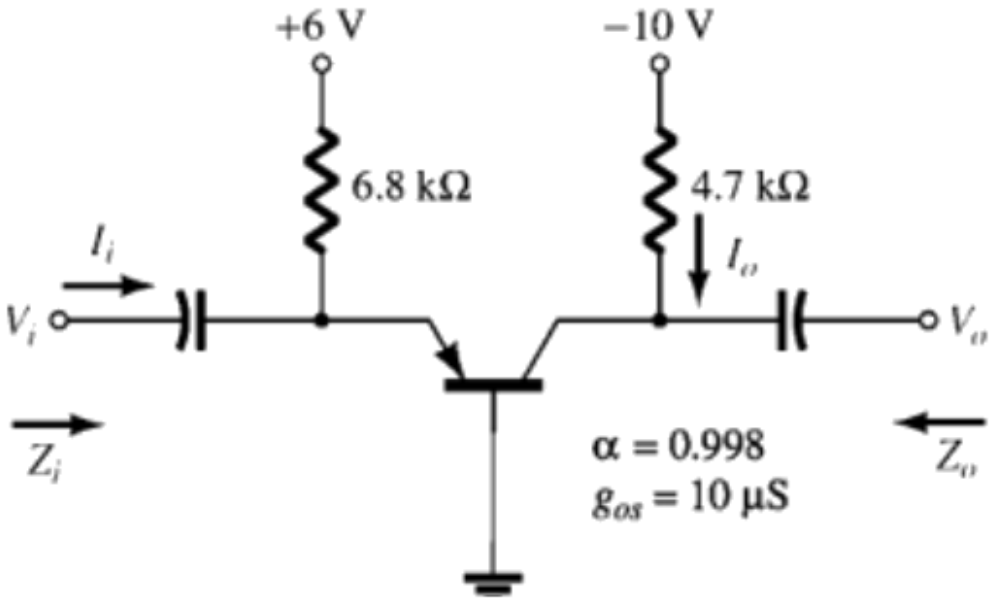
1. Determine r_e.
2. Calculate V_B and V_C.
3. Calculate Z_i and $A_V = V_o/V_i$.



WS6 Q6 figure (Figure 6 of the workshop sheet)

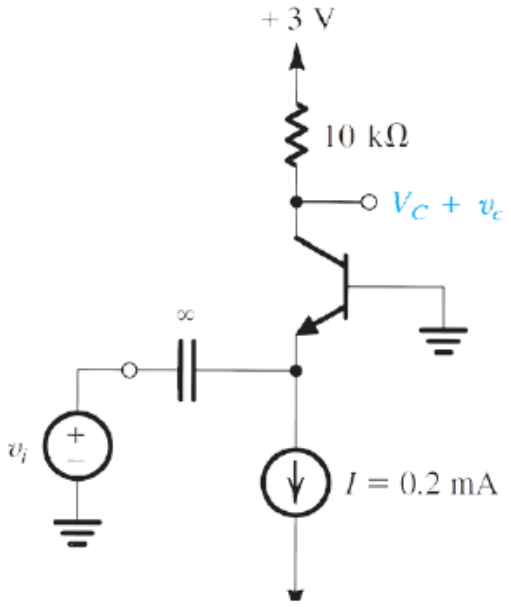
WS6 Q7 (course question). For the common-base configuration of figure 7:

1. Determine r_e.
2. Calculate Z_i and Z_o.
3. Calculate A_V.



WS6 Q7 figure (Figure 7 of the workshop sheet)

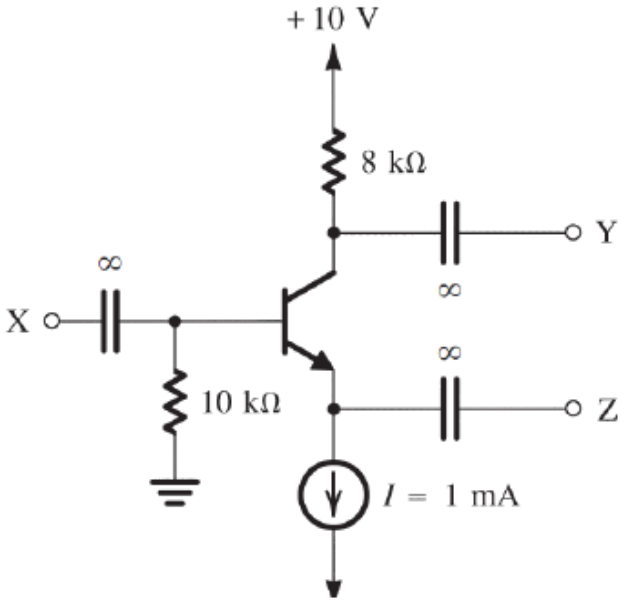
WS6 Q4 (course question). The transistor amplifier shown in the figure 9 [the figure below; the sheet labels it Figure 4] is biased with a current source $I = 0.2$ mA and has a very high β. Assuming the transistor is in the active state, the DC voltage of the collector is 1 V, and $r_e = 125\ \Omega$. Replace the transistor with the **T model** and find voltage gain, $\frac{v_c}{v_i}$.



WS6 Q4 figure (Figure 4 of the workshop sheet)

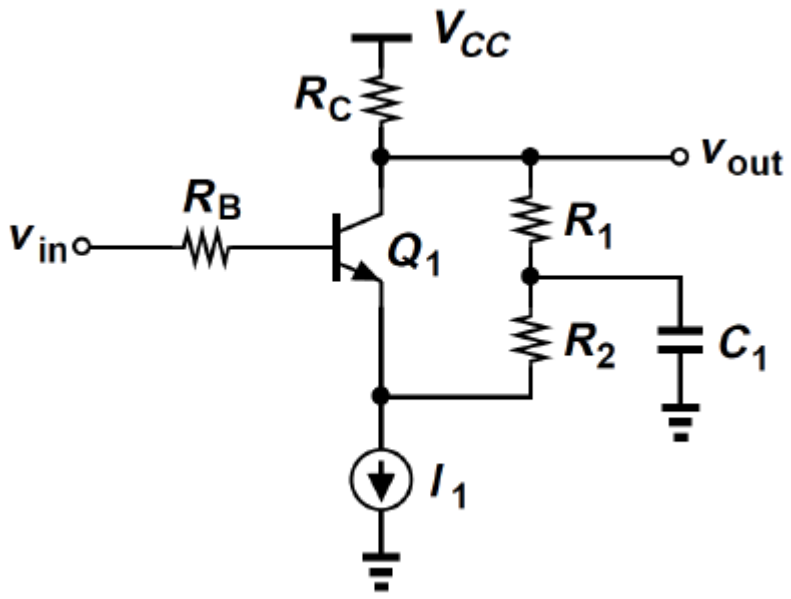
WS6 Q5 (course question). The transistor amplifier shown in the figure 10 [the figure below; the sheet labels it Figure 5] is biased with a current source $I = 1$ mA and has $\beta = 100$ and $V_A = 100$ V. Assuming that $g_m = 39.604$ mA/V, $r_\pi = 2.525\ \text{k}\Omega$ and $r_o = 101\ \text{k}\Omega$, answer the following questions:

1. If terminal Z is connected to ground, X to a signal source v_{sig} with a source resistance $R_{sig} = 2\ \text{k}\Omega$, and Y to an 8-kΩ load resistance, use the hybrid-π model to draw the small-signal equivalent circuit of the amplifier.
2. Calculate the overall voltage gain, $\dfrac{v_y}{v_{sig}}$.



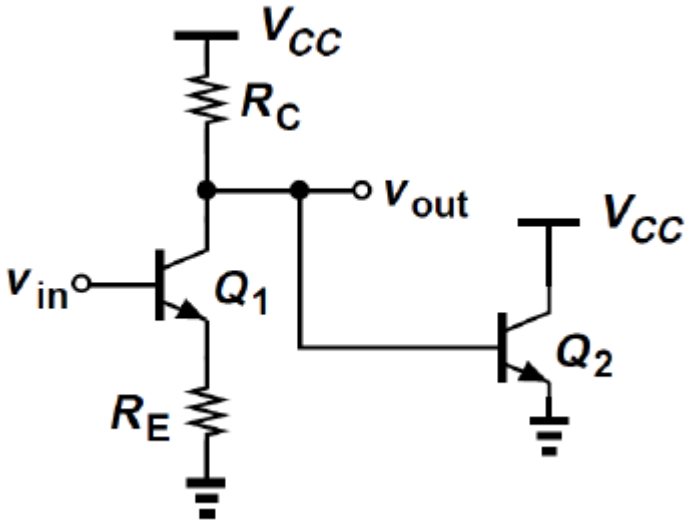
WS6 Q5 figure (Figure 5 of the workshop sheet)

WS6 Q9 (course question). The transistor amplifier shown in the figure 9 is biased with a current source I_1. Using a hybrid-π model, find expressions for AC voltage gain ($A_v = \dfrac{v_{out}}{v_{in}}$), input and output impedance. Neglect early voltage effect.



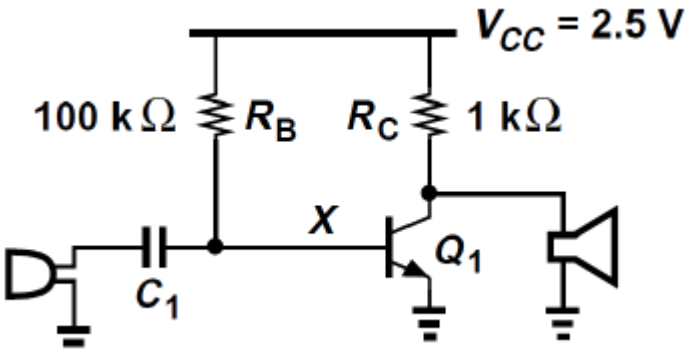
WS6 Q9 figure (Figure 9 of the workshop sheet)

WS6 Q10 (course question). The two BJTs in the amplifier of figure 10 are different. Using hybrid-π models, find an expression for AC voltage gain ($A_v = \frac{v_{out}}{v_{in}}$) of the amplifier.



WS6 Q10 figure (Figure 10 of the workshop sheet)

WS6 Q11 (course question). Typical speakers incorporate a solenoid (inductor) to actuate a membrane. The solenoid exhibits a very low DC resistance, e.g., less than 1 Ω. Having learned about AC coupling, the student in the previous Example modifies the design to that shown in figure 11 and attempts to drive a speaker. Unfortunately, the circuit still fails. Explain why. How can we resolve the issue?

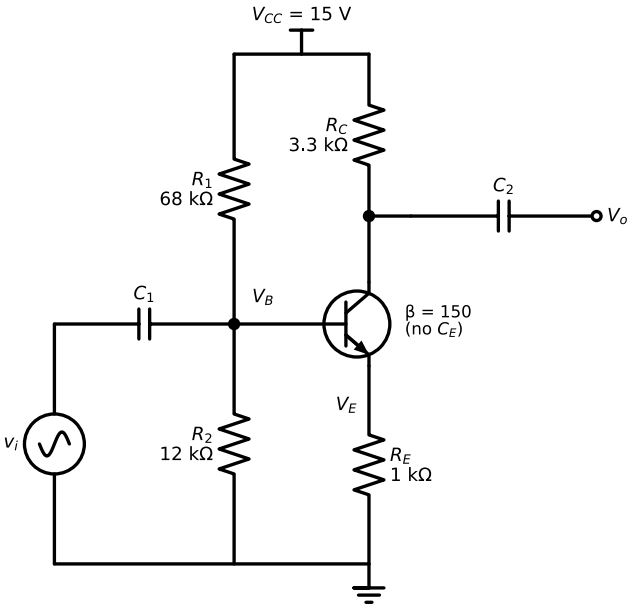


WS6 Q11 figure (Figure 11 of the workshop sheet)

New questions

N1 (easy) [5 marks]. A voltage-divider CE amplifier has $V_{CC} = 15$ V, $R_1 = 68\ \text{k}\Omega$, $R_2 = 12\ \text{k}\Omega$, $R_C = 3.3\ \text{k}\Omega$, $R_E = 1\ \text{k}\Omega$ (no bypass capacitor), $\beta = 150$.

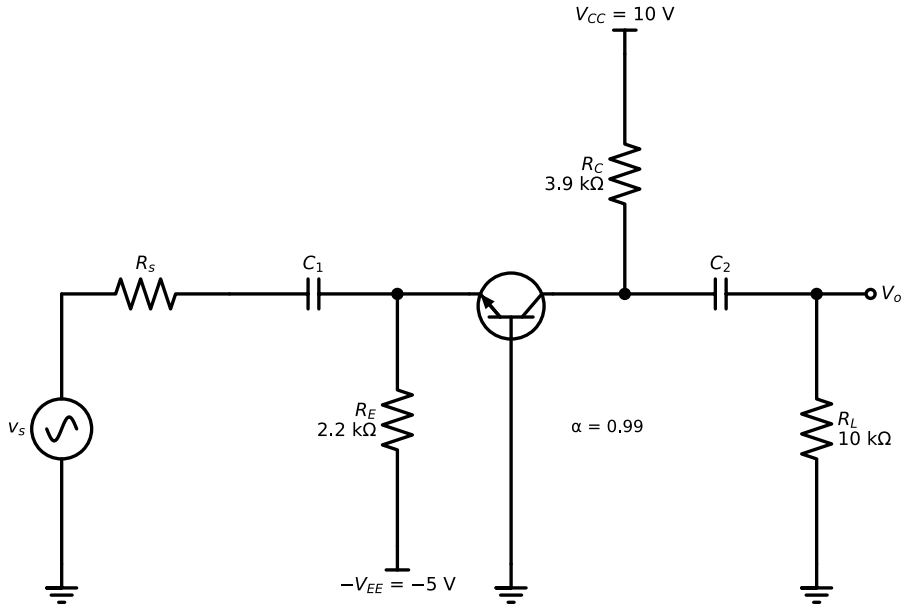
a. Find the Q-point using the approximate method, after checking that it is allowed. [2]
b. Find r_e, Z_i, Z_o and A_v. Compare A_v with $-R_C/R_E$ and with the gain if R_E were bypassed. [3]



N1: voltage-divider CE with R_E unbypassed

N2 (medium) [7 marks]. A common-base stage has $V_{EE} = 5$ V through $R_E = 2.2\ \text{k}\Omega$ to the emitter, $V_{CC} = 10$ V through $R_C = 3.9\ \text{k}\Omega$ to the collector, base grounded, $\alpha = 0.99$. A load $R_L = 10\ \text{k}\Omega$ is coupled to the collector.

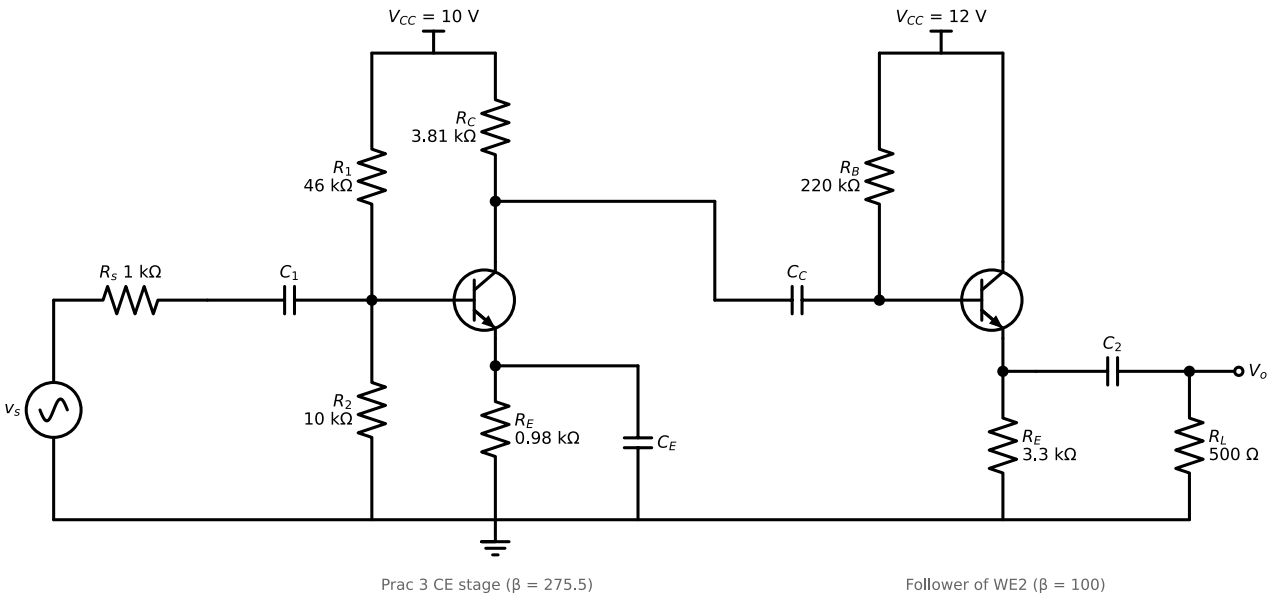
a. Find I_E, r_e and V_{CB}. [2]
b. Find Z_i, Z_o, $A_{v_{NL}}$ and A_{v_L}. [3]
c. Find A_{v_s} for $R_s = 50\ \Omega$ and for $R_s = 1\ \text{k}\Omega$. Comment. [2]



N2: common-base stage, signal into the emitter through C_1, base grounded

N3 (medium–hard, multi-part) [10 marks]. Your Prac 3 amplifier (bypassed; $Z_i = 3.60$ kΩ, $Z_o = 3.81$ kΩ, $A_{v_{NL}} = -164$, from Worked example 4) is driven from $R_s = 1$ kΩ and must drive $R_L = 500\ \Omega$.

a. Using the two-port approach, find A_{v_L} and A_{v_s} with the load connected directly. [3]
b. The emitter follower of Worked example 2 ($R_B = 220$ kΩ, $R_E = 3.3$ kΩ, β = 100, $r_e = 12.60\ \Omega$) is inserted between the CE collector and the 500 Ω load. Find the follower's Z_i with the 500 Ω load. [2]
c. Find the CE stage gain now that it is loaded by that Z_i, the follower gain, and the overall A_{v_s}. [3]
d. Find the output impedance the 500 Ω load now sees. [2]



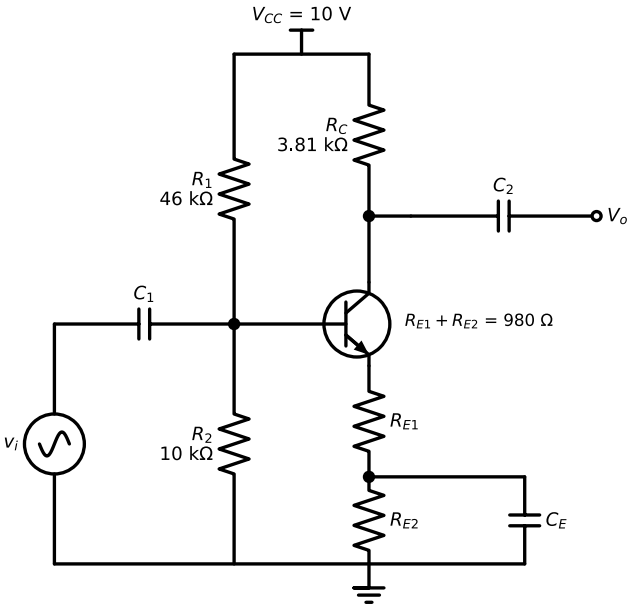
N3: the Prac 3 CE stage driving the emitter follower of Worked example 2, which drives the 500 Ω load

N4 (design) [6 marks]. In the Prac 3 amplifier ($I_E = 1.12$ mA, $r_e = 23.2\ \Omega$, $R_C = 3.81$ kΩ, $R' = 8.214$ kΩ, β = 275.5) you want a no-load gain of -10 without changing the Q-point.

a. Explain how to split $R_E = 980\ \Omega$ and choose the unbypassed part R_{E1}. Pick standard values. [3]

b. Find A_v and Z_i with your values. Repeat Z_i for β = 100. [2]

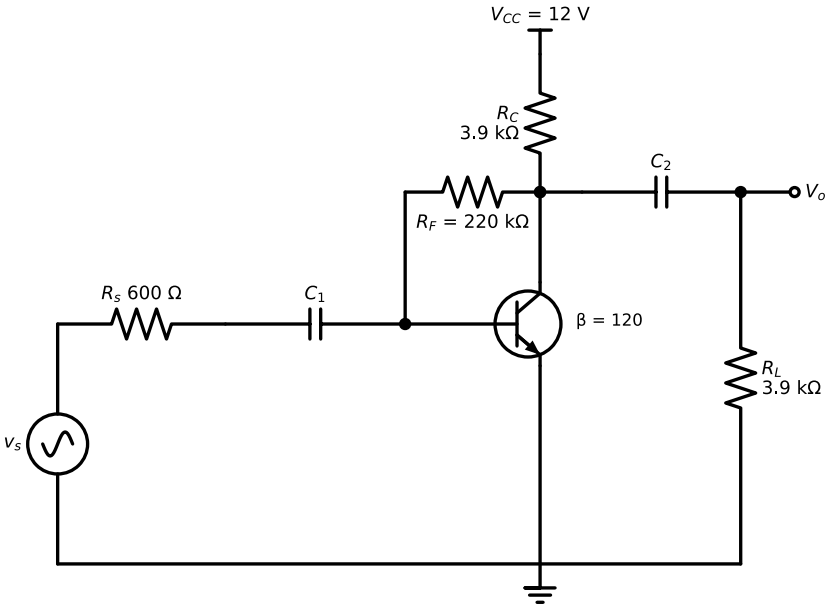
c. I_E rises 10 % with temperature. By what percentage does A_v change, bypassed versus your design? [1]



N4: the Prac 3 amplifier with R_E split into an unbypassed R_{E1} and a bypassed R_{E2}

N5 (hard) [8 marks]. The WS6 Q8 collector-feedback amplifier ($r_e = 13.08\ \Omega$, β = 120, $R_C = 3.9$ kΩ, $R_F = 220$ kΩ, ignore r_o) now drives $R_L = 3.9$ kΩ and is fed from $R_s = 600\ \Omega$.

a. Find A_{v_L}. [2]
b. Use the Miller view to find Z_i with the load connected. Explain why it differs from the unloaded 507 Ω. [3]
c. Find A_{v_s}. [1]
d. A student uses the two-port approach with the unloaded Z_i and $Z_o = R_C \parallel R_F$. What do they get, and why is it wrong? [2]



N5: the WS6 Q8 collector-feedback amplifier fed from R_s = 600 Ω and driving R_L = 3.9 kΩ

6. Quick recall check

1. A CE stage has $R_C = 4.7$ kΩ and $R_E = 1.2$ kΩ unbypassed. Roughly what is A_v, and which transistor parameter is missing from it?
2. What is Z_o of an emitter follower with $r_e = 10\ \Omega$, $R_E = 2$ kΩ, driven from $R_s = 2$ kΩ (ignore R_B), β = 100?
3. Which two configurations have Z_i that changes when R_L is connected?
4. Write A_{v_s} in terms of Z_i, R_s and A_{v_L}.
5. With $R_E = R_{E1} + R_{E2}$ and only R_{E2} bypassed, which resistance goes in the DC loop and which in A_v?